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A quantitative assessment of potential benefits and challenges of international researcher mobility for home and host countries: evidence from the Chinese Scholarship Council programmes

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Abstract
Understanding the benefits and challenges of international mobility programmes for both home and host countries is essential for evaluating them. This study provides a quantitative assessment of the Chinese government-sponsored international mobility programmes, the Chinese Scholarship Council (CSC) programmes, by examining their effects on home and host countries across disciplines from five key perspectives: output volume, impact, collaboration, funding, and involvement in security-sensitive research areas. Our findings suggest that, for the home country (China), the funding programmes enhance China's research capabilities in alignment with its strategic development priorities. Additional evidence suggests that the funding programmes facilitate international collaboration and allow awardees to work with highly cited researchers. For host countries, the funding programmes promote collaboration with China and may support host researchers where funding might be scarce. Also, the funding programmes place little emphasis on dual-use research and have limited ties to the "Seven Sons of National Defence" universities. This finding addresses current concerns in some host countries regarding security challenges. The study provides evidence for policymakers to develop strategies that maximise the benefits of international mobility programmes and address associated challenges.

Keywords Research funding · International mobility · Quantitative assessment · China Scholarship Council · Research collaboration

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Introduction

International researcher mobility has significantly increased in the context of the globalisation of science, turning the academic sector into a 'global market' of multicultural teams with high levels of temporary foreign researcher participation (Baruffaldi et al., 2020; Gureyev et al., 2020). Mobility is widely seen as a strategic tool for enhancing national research capacity or sustaining global competitiveness (Jonkers & Cruz-Castro, 2013), and many nations prioritise mobility support to boost research capabilities or maintain their scientific status (Horta et al., 2020; McManus & Nobre, 2017; Meyer, 2003; Stephan & Levin, 2001; Zahler & Menino, 2018, pp. 65–83). Notable examples include the Fulbright programme (US), the Sandwich programme (Brazil), and initiatives from the Chinese Scholarship Council (CSC) (Kahn & MacGarvie, 2016; Shen, 2018).

Despite the expansion of mobility initiatives, concerns such as brain drain, knowledge leakage, and integration challenges raise doubts on whether such mobility truly achieves the reciprocal benefits envisioned by the "circulation of talent" framework (Daugeliene & Marcinkeviciene, 2009; Jonkers & Cruz-Castro, 2013; Saxenian, 2005). In light of these concerns, a comprehensive understanding of its impact requires evaluating both the benefits and challenges for both home and host countries.

A substantial body of previous research focuses on the multifaceted benefits of international mobility, either from the home country's side or the host country's side. Host countries typically benefit through increased research output and enhanced innovation capacity (Crown et al., 2020; Jonkers & Cruz-Castro, 2013; Jonkers & Tijssen, 2008; Zhao et al., 2020), fuelled by the contributions of international researchers. Meanwhile, home countries gain from returning researchers who bring back their tacit knowledge and social capital (Baruffaldi & Landoni, 2012; Cao et al., 2020; Jonkers, 2011; Jonkers & Cruz-Castro, 2013; Jonkers & Tijssen, 2008; Shin et al., 2014). These returnees not only contribute to enhancing domestic scientific performance but also promote international collaboration, which further internationalises the domestic research system (Bolli & Schläpfer, 2015; Cao et al., 2020; Gibson & McKenzie, 2014; Smyth & Mishra, 2014).

However, this "mutual benefit" cannot be achieved in all situations. Mobile scientists may encounter substantial personal and professional barriers, including cultural differences, language barriers, and administrative constraints such as visa restrictions. These issues can hinder their full integration into host countries and, in turn, limit their contribution to the host country's research system (Crown et al., 2020; Gao & Liu, 2021; Gureyev et al., 2020; Hooghe & Martin, 2024). Moreover, in the context of growing geopolitical tensions and heightened concerns over knowledge security, international mobility may raise concerns in host countries about the unintended transfer of critical technological or sensitive scientific knowledge. From the perspective of home countries, not all mobility leads to the expected return of talent. In some cases, host countries retain highly skilled researchers on a long-term basis, fuelling concerns about brain drain, especially in nations with less developed research infrastructures. These multifaceted challenges underscore the urgent need for a more comprehensive and bilateral evaluation of international mobility.

Our study aims to fill this gap by empirically examining the CSC programmes to assess the actual benefits and risks for both home and host countries. The CSC offers a particularly relevant case in the context of current heightened geopolitical tensions. In the past two years, several countries, including Sweden (Nilsson & Sadikovic, 2023), Denmark (Schou Drivsholm, 2023), Germany (Felden, 2023; Sharma, 2023), the Netherlands (NL Times, 2023), and the United States (Zwetsloot, 2020), have reassessed or even suspended their

collaboration with CSC-funded scholars (DutchNews, 2023; NL Times, 2023; Sharma, 2023). These decisions have been largely driven by concerns that CSC-funded academic mobility may serve as a channel for the unintended transfer of sensitive research back to China, particularly in strategic or high-tech research areas. Such concerns are especially pronounced in relation to scholars affiliated with the so-called "Seven Sons of National Defence", a group of Chinese universities with close ties to the military. Against this backdrop, our study contributes timely and valuable insights by systematically evaluating multifaceted impacts from both the home and host countries.

Our study assesses the benefits and challenges of the CSC programmes for China and six major host countries, using data from over 100,000 scientific publications acknowledging CSC funding between 2009 and 2021. The benefits and challenges are assessed across five key perspectives: output volume, impact, collaboration, funding, and security sensitivity and involvement in security-sensitive research areas. We compare CSC-funded outputs with those of researchers in both China and the host countries, aiming to address the following research questions:

1. How do CSC-funded publications compare in number to other publications from the home and host countries across disciplines?
2. How does the citation impact of publications by CSC-funded researchers compare to other publications in the host countries across disciplines?
3. What are the supported researchers' collaboration patterns across disciplines?
4. Do CSC-funded researchers work in areas that are relatively underfunded?
5. To what extent is CSC-funded research associated with sensitive areas of research in the host countries?

The structure of this paper is as follows: "Research background" section provides an overview of CSC programmes. "Data and methods" section describes the methodology of our study. We then present and discuss the findings in "Results" section. "Discussion and conclusion" section summarises our key conclusions. Finally, "Limitation and future research" section discusses potential directions for future research.

Research background

Overview of the China Scholarship Council programmes

The China Scholarship Council (hereinafter CSC), officially the "National Education Fund Management Committee" (Fedasiuk, 2020), is a non-profit organisation established by the Chinese Ministry of Education in 1996. The CSC aims to select talented undergraduate students, graduate students, postdoctoral fellows, and visiting scholars to study abroad, with the objective of facilitating academic collaboration and international experience (China Scholarship Council, 2017).

The CSC administers a comprehensive portfolio of international funding programmes (China Scholarship Council, 2025). These initiatives include: (i) Senior Research Scholars and Visiting Scholar Programme, (ii) National High-Level University Postgraduate Programme, (iii) Short-Term Overseas Exchange Programme for PhD Supervisors, (iv) University Cooperation Programme (Young Faculty Overseas Training Programme), (v) Western Region Talent Development Special Programme & Local Cooperation Programme,

(vi) International Organisation Talent Development Programme, (vii) National Regional Studies & High-Level Foreign Language Talent Development Programme, (viii) Special Programme for the Cultivation of Artistic Talent, (ix) Rural Revitalisation Talent Development Programme, and (x) International Cooperation Programmes (e.g., CSC–Cambridge, CSC–DAAD, CSC–KTH).

The duration of grants varies by academic status and programme type. Each programme supports specific categories of recipients—ranging from undergraduate students to senior researchers—with funding durations aligned to their academic status. Table 1 provides an overview of the main programmes, their eligible participants, and the corresponding funding periods.

CSC recipients receive financial support to undertake a stay abroad at an institution of their own choosing. The funding typically covers a one-time round-trip international airfare and a stipend for the duration of the funding period. This stipend is intended to cover expenses such as food, accommodation, registration fees, bench fees, transport, telephone charges, books and materials, medical insurance, social activities, a one-off resettlement allowance, visa extension fees, personal allowances, service fees, and academic activity subsidies. However, tuition fees are not included. The programmes require awardees to return to China and work there for two years after completing their period of overseas study or research (Gu et al., 2024; Wang et al., 2024). In addition, joint PhD, Master's, and undergraduate exchange students are required to undertake their overseas studies as part of their degree programmes. Upon completion, they must return to China to defend their dissertation and fulfil all degree requirements.

In addition, the CSC encourages recipients to acknowledge the support of the CSC in research projects, journal articles, or any other scientific outputs related to the funded study by stating: "This research/project/publication was supported by the China Scholarship Council (CSC).

The evaluation of applicants is centralised in Beijing. The CSC is responsible for the scientific assessment of all applications. Each review panel consists of field experts appointed by the CSC. The evaluation process adheres to the principles of openness, fairness, and impartiality, following the procedure of "individual application, institutional recommendation, expert review, and merit-based admission." Each application is assessed based on: (i) ethical conduct and academic integrity, (ii) the applicant's overall qualifications, including academic background, scientific achievements (publications, patents, awards), international communication skills (including language proficiency), and career prospects, (iii) the quality, originality, and feasibility of the research project, (iv) the level of the applicant's home institution, and (v) the level and capacity of the proposed host institution and its alignment with the applicant's research interests.

With regard to this final aspect, applicants are expected to carefully select host institutions that best align with the objectives of their research proposal. They must also obtain invitation letters from the host institutions prior to applying for CSC funding.

Previous research on the China Scholarship Council

The CSC programmes have received limited attention in the scientific literature (Wang et al., 2024). Previous research on the CSC primarily focuses on its role in knowledge production, research collaboration, and the academic development of individual recipients (Gu et al., 2024; Horta et al., 2020; Liu et al., 2021; Shen, 2018; Wang et al., 2024). For example, Gu et al. (2024) examined the impact of CSC support on scientific collaboration

Table 1 Overview of CSC funding programmes by recipient type and duration

Programmes	Targets	Duration
Senior Research Scholars and Visiting Scholar Programme	Senior researchers, Visiting scholars	3–6 months (senior researchers); 3–12 months (visiting scholars)
National High-Level University Postgraduate Programme	Full-time PhDs, Joint PhDs	36–48 months (full-time PhDs); 6–24 months (joint PhDs)
Short-Term Overseas Exchange Programme for PhD Supervisors	PhD supervisors	3–6 months
University Cooperation Programme (Young Faculty Overseas Training Programme)	Visiting scholars, Postdocs	3–12 months (visiting scholars); 6–24 months (postdocs)
Western Region Talent Development Special Programme & Local Cooperation Programme	Senior scholars, Visiting scholars, Postdocs	3–6 months (senior scholars); 3–12 months (visiting scholars); 6–24 months (postdocs)
International Organisation Talent Development Programme	Interns, Visiting experts	3–12 months
National Regional Studies & High-Level Foreign Language Talent Development Programme	Senior researchers, Visiting scholars, Postdocs, Full-time PhDs, Joint PhDs / masters	3–6 months (senior researchers); 3–12 months (visiting scholars, joint masters); 6–24 months (joint PhDs, postdocs); 36–48 months (full-time PhDs)
Special Programme for the Cultivation of Artistic Talent	Visiting scholars, Postdocs, Joint PhDs / masters	3–12 months (Visiting scholars, Joint masters); 6–24 months (Postdocs, Joint PhDs)
Rural Revitalization Talent Development Programme	Visiting scholars, Full-time masters, Joint masters	3–12 months (visiting scholars, joint masters); 12–24 months (full-time masters)
International Cooperation Programmes (e.g., CSC—Cambridge, CSC—DAAD, CSC—KTH)	Postdocs, Joint PhDs	6–24 months

and productivity by comparing CSC-funded visiting PhD scholars with their non-mobile counterparts. Similarly, Wang et al. (2024) conducted a long-term, systematic evaluation of CSC's performance and impacts in terms of citation impact, publication output, and collaboration patterns. Shen (2018) and Shen et al. (2017) explored the benefits of the CSC from an individual perspective, highlighting its role in enhancing recipients' academic skills.

The CSC also has implications for host countries. These countries have shown increasing interest in the broader impact of international sponsorship programmes, especially in light of recent geopolitical tensions. Questions have been raised regarding whether such initiatives generate truly mutual benefits. However, existing research on the CSC pays relatively little attention to the host country perspective. This oversight is not unique to the CSC. As discussed in the introduction, current studies on international funding programmes, and international mobility more broadly, offer limited systematic and bilateral evaluations encompassing both home and host countries. To address this gap, our study undertakes a comprehensive assessment of the CSC from both perspectives.

Data and methods

Data source

We rely on the funding acknowledgements in scientific publications to identify publications reporting research funded by the CSC and also the specific researchers who received this funding. The Web of Science (WoS) began incorporating funding acknowledgements for publications in the Science Citation Index in 2008, and since 2015 also for those covered in the Social Sciences Citation Index (Paul-Hus et al., 2016).

In November 2023, we retrieved from the CWTS in-house version of WoS a total of 104,148 publications from the period 2009–2021 acknowledging financial support from the CSC. We considered all name variants of the CSC appearing in the acknowledgements (e.g. 'the CSC Scholarship' and 'the Chinese Scholarship Council') as captured in the CWTS in-house database of funding organisations (Van Honk et al., 2016). Our search was restricted to articles (90.9%), reviews (3.4%) and proceedings papers (5.6%).

We classified these publications into five scientific fields by drawing on the CWTS classification system. This system uses the citation cluster method proposed by Waltman and Van Eck (2012) together with the Leiden algorithm for network clustering (Traag et al., 2019). Each publication is assigned to a single research area at the micro level of the classification system based on citation relations, so that each research area gathers publications dealing with similar topics (Waltman & Van Eck, 2012). The CWTS classification includes five broader fields: Physical Sciences and Engineering, Mathematics and Computer Science, Biomedical and Health Sciences, Life and Earth Sciences, and Social Sciences and Humanities.

Identifying CSC awardees and their host countries

To determine the countries hosting CSC awardees, we first identified which of the 604,347 authors in the 104,148 publications acknowledged funding from the CSC. This allowed us to determine the countries where these authors are based, through the institutional affiliations indicated in the publications.

Our strategy to identify CSC awardees consisted of two steps. In the first step, for all publications with a single author, we assumed this author was supported by the CSC. For publications with multiple authors, we searched for the authors' names in the sentence of the funding acknowledgement where the CSC was mentioned. To do so, we first extracted from the acknowledgement the full sentences mentioning the CSC.

We then searched for the authors' names within these sentences. For this purpose, we created several variants for each author (e.g. full name and last name, initials and surnames, etc.). Furthermore, because in some cases the order of the author in the byline is mentioned rather than the author's name, we also considered ordinal numbers based on the author byline, such as "the first author" and "the second author".

We manually examined the sentences in which several author names appear together with the CSC mention, to ensure the correct identification of the specific author awarded with a CSC grant. In this first step, we were able to identify CSC scholarship awardees with a high degree of certainty. For example, in one publication with four authors, we determined which of them (Jian Wang) received funding from the CSC: "This work was supported by grant of the DFG-Hu 729/10 and DFG-Hu 729/13 to R.H. and by the China Scholarship Council to J.W.". To reach this conclusion, we had previously created multiple variants for each author, such as "Jian Wang", "Wang Jian", "J Wang", "J. Wang", "J. W.", "J.W.", and "the first author", among others. Similarly, for "Ruijin Huang" we used variants such as "Huang Ruijin", "R Huang", "R. Huang", "R.J. Huang", "R. H.", and "the first author". In this example, the initials "J.W." and "R.H." were found in the same sentence where the CSC is mentioned We further manually examined the sentence and identified "Jian Wang" as the CSC-funded scholar.

In the second step, we addressed all publications for which we were unable to identify any CSC awardee in step 1. For this subset of publications, we first retrieved all the publications produced by each CSC awardee identified in step 1, using a large-scale author name disambiguation algorithm developed by Caron and van Eck (2014).

This algorithm handles homonymy and name variants to identify all scientific publications produced by a given researcher. We then crossed the complete list of publications of these previously identified CSC awardees with the subset of CSC-funded publications for which we were not able to identify any CSC awardee. To ensure, as far as possible, that the publication was related to the CSC scholarship, we only considered a publication to be CSC-funded for a given researcher if its publication year fell between the first and the last publication identified in step 1 for that researcher, with a maximum period of three years.

This second step represents a somewhat looser but reasonable criterion in the assignment of publications to CSC awardees. The combination of steps 1 and 2 allowed us to identify CSC awardees for 55.16% (57,449) of publications acknowledging CSC funding. Table 2 shows the number of publications for which CSC awardees were identified in each step.

Next, we identified host countries based on the institutional affiliations indicated in these publications to determine the countries to which the CSC-funded researchers moved. We used full counting to assign publications to countries; that is, we assigned one full publication to each country involved in a publication. Our study focuses on the top six countries with the highest number of publications produced by CSC-funded researchers (excluding China). These countries are: the United States, Germany, the United Kingdom, Australia, France and the Netherlands.

Table 2 Strategy to identify CSC awardees among authors

Search strategy	Publications
Step 1.1—Sole author	968
Step 1.2—Publications with more than one author, identified in the same sentence of the funding acknowledgement as the CSC	55,148
Step 2—Publications by authors identified in Steps 1.1 and 1.2 in a given time period but not mentioned in the funding acknowledgements	1333

Dual-use research

Our strategy to identify research areas potentially related to dual-use technologies builds on the previous efforts undertaken by the Joint Research Centre (JRC) of the European Commission, through a monitoring system called 'Tools for Innovation Monitoring' (TIM). More specifically, we relied on the TIM Dual-Use Web Platform, developed by JRC and the University of Liège (European Commission, 2024), which also includes a dashboard.[1]

The TIM Dual-Use Web Platform considers the dual-use items listed and described in Regulation (EU) 2021/821, as amended by the Commission Delegated Regulation (EU) 2023/2616, as well as several emerging technologies not listed in these regulations but with potential dual-use applications such as artificial intelligence, blockchain or cyber-surveillance, among others. These regulations organise the dual-use items into ten categories, covering different domains: Category 0: Nuclear materials, facilities and equipment; Category 1: Pathogens; Category 1_1: Special materials and related equipment; Category 2: Materials processing; Category 3: Electronics; Category 4: Computers; Category 5: Telecommunications and information security; Category 6: Sensors and lasers; Category 7: Navigation and avionics; Category 8: Marine; Category 9: Aerospace and propulsion.

For the above ten categories and for emerging technologies of potential concern, TIM, with the assistance of domain experts, tailored respective dual-use search queries. These queries were constructed based on keywords associated with the EU dual-use control list, as well as emerging technologies and their scientific and technical synonyms. Additionally, domain experts helped optimise keyword selection (DU related) and designing Boolean operators (AND, OR, NOT) to maximise the results while striving for the best possible relevance. Finally, search queries were applied to the Scopus database, the European Patent Office's PATSTAT database, and the EU-funded projects database (CORDIS) to retrieve relevant documents.[2] The retrieved documents represent research outputs relevant to dual-use technologies. For example, research on the *Ebola virus* falls under Category 1: Pathogens, with the following search query: *topic:* ((ebola AND virus AND ("synthetic biology" OR GMO OR "Genetically Modified Organism" OR PCR OR "Polymerase Chain Reaction" OR genom* OR genetics OR replication OR sequenc* OR RNA OR "ribonucleic acid" OR isolat*)) NOT (epidemiology AND outbreak)). A search conducted in 2025 yielded 2831 documents, including journal articles, EU-funded projects, book chapters, conference papers, patents, and review articles.[3]

[1] https://www.timanalytics.eu/TimTechPublic/dashboard/index.jsp#/space/s_1536.
[2] https://knowledge4policy.ec.europa.eu/text-mining/tim-dual-use_en.
[3] https://www.timanalytics.eu/TimDualUse/dashboard/index.jsp#/space/s_14?ds=89.

In our study, we focus on these ten categories. We used these tailored search strategies to retrieve the relevant publications in WoS. The publications retrieved with these search strategies are considered an indication of potential research activity related to dual-use research, but they are not considered an exhaustive or highly precise identification of all dual-use research for a number of reasons. First, because not all dual-use research is published in scientific journals, and also because researchers may use terminology not captured by the search strategies.

CSC-funded research: home versus host country perspectives

Extensive research shows that international mobility generates mutual benefits for both home and host countries by fostering international academic collaboration (Gu et al., 2024; Wang et al., 2019, 2024), increasing number of publications (Cao et al., 2020) and improving citation impact of research (Jing et al., 2023; Jonker & Cruz-Catro, 2013). As a prominent example of government-supported international mobility initiatives, the CSC programme offers an ideal case for examining how these benefits materialise. In this study, we assess whether CSC-funded researchers demonstrate higher research performance—measured by the publications they produce and their citation impact—and stronger academic collaboration compared to their peers across disciplines.

Additionally, Hooghe and Martin (2024) argue that collaboration with CSC-funded researchers may help Dutch researchers to pursue types of research that might be difficult to finance through other funding sources in specific research areas. Building on this, we examine the contributions of CSC-funded researchers across different research areas, with particular attention to the funding patterns of their work in host countries.

Lastly, in the context of rising geopolitical tensions, several host countries have raised concerns about potential risks to national intellectual security associated with CSC-funded projects (Nilsson & Sadikovic, 2023; Schou Drivsholm, 2023; Sharma, 2023; Felden, 2023; NL Times, 2023; Zwetsloot, 2020). These concerns are particularly pronounced in cases involving researchers affiliated with the "Seven Sons of National Defence" or those engaged in research related to dual-use technologies. In this study, we examine the extent to which CSC-funded researchers and their outputs are relevant to these security-related concerns.

We rely on a number of bibliometric indicators to analyse the five dimensions mentioned above: Number of publications, citation impact, collaboration patterns, funding patterns, and security-sensitive research. In our study, we mainly rely on size-independent indicators to facilitate comparison across countries and disciplines. The indicators are summarised in Table 3.

As basic indicators, we use P and PP (discipline) which refer to the number of publications produced. P represents the number of CSC-funded publications, while PP (discipline) illustrates the proportion of CSC-funded publications in each discipline.

In addition, MNCS and PP (top 10%) are frequently used as a proxy of scientific impact. Both are normalised indicators that account for differences in the number of citations received due to publication year and scientific discipline (Waltman & Van Eck, 2012). This normalisation enables the comparison of the citation scores of publications produced in different years and different scientific fields.

The MNCS represents the average number of citations received by a set of publications, while the PP (top 10%) indicates the proportion of publications in the top 10% of most cited worldwide. Because the MNCS indicator is sensitive to outliers, we use it in

Table 3 Overview of selected bibliometric indicators

Dimension	Indicator	Remarks
Number of publications	P	Number of CSC-funded publications (based on fractional counting) in the host country
	PP (discipline)	Proportion of CSC-funded publications in each discipline, within the host country
Citation impact	MNCS	Mean Normalised Citation Score: average number of citations received by publications from a country, normalised by field (CWTS classification) and publication year
	PP (top 10%)	Proportion of publications that rank among the top 10% most frequently cited publications, compared with others in the same field and year
Collaboration Pattern	PP (No collab)	Proportion of publications produced by a single institution
	PP (nat collab):	Proportion of publications co-authored by multiple institutions within the same country affiliated with CSC awardees
	PP (int collab)	Proportion of publications co-authored by institutions in two or more countries, affiliated with CSC awardees
	AVG (authors)	Average number of authors per article
Funding patterns	PP (funded)	Percentage of publications by the number of funders acknowledged
Security-sensitive research	PP (SSND): Share of publications with "Seven Sons of National Defence"	Proportion of publications co-authored with institutions from the "Seven Sons of National Defence"
	PP (DUT): Share of dual-use technology publications	Proportion of CSC-funded publications in research fields associated with dual-use technologies

combination with PP (top 10%) to provide a more robust estimate of scientific impact. MNCS and PP (top 10%) are calculated using a variable-length citation window, meaning citations are counted from the publication date of each paper up to the moment the indicators are computed. We excluded self-citations.

To assess scientific collaborations, we analyse co-authorship, a common approach in bibliometrics (Glänzel, 2003, P73.). PP (nat collab) denotes the propensity of CSC awardees to engage in domestic collaborations (i.e. when a publication is produced by two or more organisations in the same country), while PP (int collab) indicates the proportion of publications produced through international collaboration, that is, by two or more countries. PP (no collab) reflects either intra-institutional collaboration or individual authorship. AVG (authors) represents the average number of authors per article. Together, these indicators show the extent to which CSC-funded researchers engage in various types of collaboration.

PP (funded) reflects percentage of publications by the number of funders acknowledged. PP (funded) provides a rough indication of whether a particular research field or country benefits from relatively more or less funding, based on the number of acknowledged funding bodies. Similarly, the indicator allows us to detect whether CSC-funded researchers work in areas that also typically receive support from other funding organisations other than the CSC, or if CSC funding appears to be the sole source of funding acknowledged in the publication. This is a coarse approximation, as the number of acknowledged funders may or may not correlate with the actual amount of funding, and the number of funders mentioned can vary significantly across countries and scientific fields. Nevertheless, we use this information to analyse whether CSC-funded researchers represent an opportunity for host organisations to leverage their expertise in areas that may not receive otherwise project-based funding.

We also calculated the proportion of publications in which the CSC-funded researcher is affiliated with one of the "Seven Sons of National Defence" universities, denoted as PP (SSND). PP (SSND) captures the connections between CSC-funded research and this group of universities. Similarly, PP (DUT) shows the proportion of CSC-funded publications that fall into dual-use research categories. By analysing these indicators, we assess the extent of CSC-funded research activity in security-sensitive areas.

Results

Overview of research output

Between the period 2009–2021, CSC grant recipients produced 104,148 publications. As shown in Fig. 1, the number of CSC-funded publications steadily increased over the years, peaking in 2020 with 16,385 publications. After this peak, the number of publications declined in 2021. Although neither the global publication output nor China's total publications count provides an ideal benchmark for CSC awardees, the comparison suggests that the drop in CSC-funded publications in 2021 may have been particularly influenced by the travel restrictions during the COVID-19 pandemic, especially when compared to trends in China and globally. In addition to pandemic-related disruptions, raising geopolitical tensions and increasing restriction on international mobility—especially between China and some countries like Canada and the US—may have further contributed to this decline (Mok et al., 2024; Song & Li, 2022; Trilokekar et al., 2021).

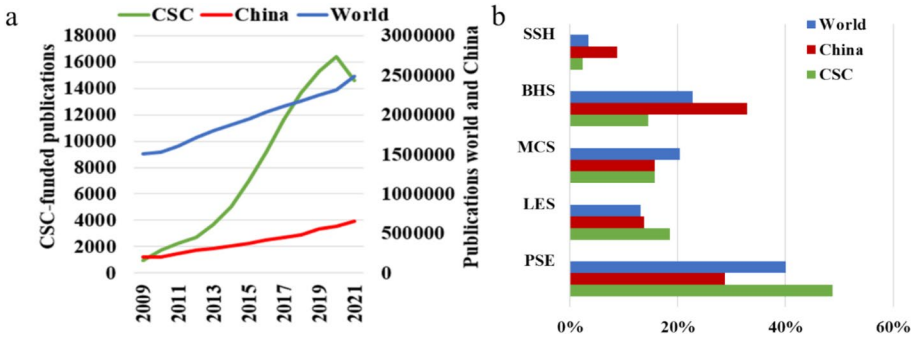

Fig. 1 Publication distribution across global, Chinese, and CSC-funded research. **a** Annual number of publications in global, Chinese, and CSC-funded research. **b** Percentage of publications by field in global, Chinese, and CSC-funded research (*BHS* Biomedical and Health Sciences, *LES* Life and Earth Sciences, *MCS* Mathematics and Computer Science, *PSE* Physical Sciences and Engineering, *SSH* Social Sciences and Humanities)

CSC awardees are highly concentrated in two scientific fields, which together account for approximately 70% of all publications: Physical Sciences and Engineering (49%) and Life and Earth Sciences (19%). The percentage of publications by CSC awardees in these two areas surpasses that of China and the world. However, while Biomedical and Health Sciences represent a major area of Chinese research activity, accounting for slightly more than one-third of all Chinese publications, the contribution of CSC awardees to this field is smaller (14%), and also lower than the global share (23%). Moreover, CSC and Chinese researchers have made modest contributions in the field of Mathematics and Computer Science, accounting for a similar share (16%), which is slightly below the global percentage (20%). The Social Sciences and Humanities account for only 2% of all CSC-funded publications, a markedly lower share compared to China's 9%.

Based on the 57,449 publications in which the CSC-funded author could be identified, we found that CSC researchers moved from China to 68 different countries. This analysis focuses on the top six destination countries hosting the largest number of CSC-funded researchers. The United States received the highest number of CSC-funded researchers (10,489), more than twice the number received by Germany (4620). Following the United States and Germany, the United Kingdom (3165), Australia (2328), France (2271), and the Netherlands (2058) were also major destinations for CSC-funded researchers.

The yearly number of publications produced by CSC-funded researchers in each of these six countries is shown in Fig. 2. A relatively sustained increase in publications can be observed in every country throughout most of the period. However, a sharp decline in publications in the US after 2019 may have been partially influenced by travel bans that prevented CSC awardees from moving to the country. In recent years, some European countries (e.g., the UK, Germany, France, and the Netherlands) have shown an increase in the number of publications. These trends suggest that CSC awardees opted to conduct their research in European countries due to difficulties in relocating to the US.

The research output of CSC awardees in these host countries is predominantly concentrated in three specific fields, which together account for over 80% of all scientific publications. Notably, approximately 50% of CSC awardees' publications fall within the Physical Sciences and Engineering field, except in the Netherlands, where this figure is 33%.

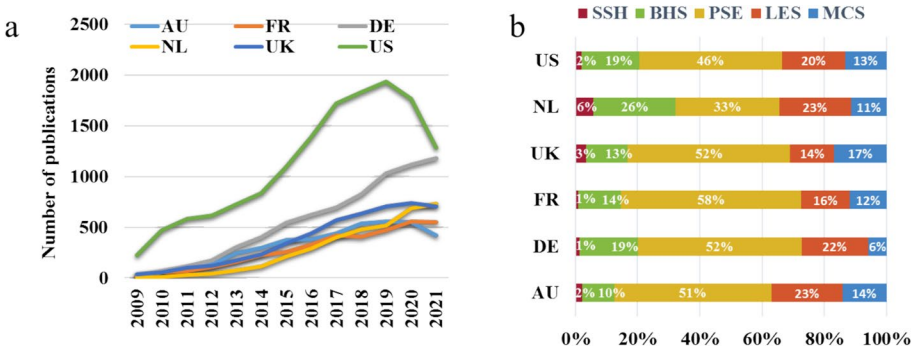

Fig. 2 Distribution of CSC-funded publications across host countries. **a** Annual number of CSC-funded publications by host country over time. **b** Percentage distribution of CSC-funded publications by field within host countries (*BHS* Biomedical and Health Sciences, *LES* Life and Earth Sciences, *MCS* Mathematics and Computer Science, *PSE* Physical Sciences and Engineering, *SSH* Social Sciences and Humanities)

Additionally, the Social Sciences and Humanities and Mathematics and Computer Science fields together constitute less than 20% of all publications. Representation in these two fields is consistently low across all host countries.

Citation impact

Figures 3 and 4 present scatter plots illustrating the citation impact of various countries by field. In each plot, the x-axis shows the Mean Normalised Citation Score (MNCS) and the proportion of publications among the top 10% most cited worldwide (PP (top 10%)) for each host country, excluding CSC-funded publications. The y-axis shows the MNCS and PP (top 10%) of CSC-funded publications within the same country. The red diagonal line (y = x) represents where the MNCS and PP (top 10%) values of CSC-funded and non-CSC-funded publications are equal. Data points above this line indicate a higher citation impact for CSC-funded publications, while points below indicate a lower impact compared to non-CSC-funded publications. The worldwide average citation impact equals 1, and 10% serves as the reference value for the PP (top 10%) indicator.

The citation impact of CSC awardees is higher than both the global average and that of Chinese scholars across all fields. This higher citation impact may partially result from CSC awardees' collaborations with host-country researchers, which enhance their international visibility and facilitate broader dissemination of their work, leading to increased citation rates. In contrast, Chinese scientists who remain at domestic institutions often face limited access to international academic networks and reduced visibility (Qiu et al., 2025), factors that may partially explain their comparatively lower citation impact.

Across the six host countries, the citation impact of CSC awardees shows varied patterns across both countries and scientific fields. In the field of Physical Sciences and Engineering, the citation impact of publications by CSC awardees is significantly higher than that of local researchers in all host countries. The higher citation impact of CSC awardees' publications may suggest that they are collaborating with leading international researchers or institutions, resulting in high-impact research outputs.

The work of CSC awardees in the US demonstrates comparable or higher citation impact across all fields. Conversely, in some countries, publications without the involvement of

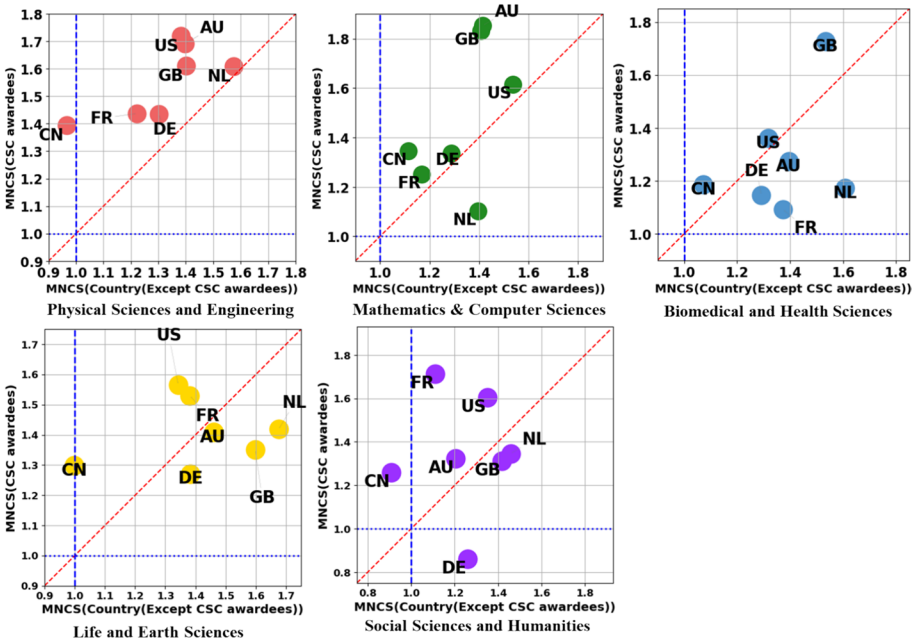

Fig. 3 Mean Normalised Citation Score MNCS of CSC recipients in their home and host countries. *Note*: The worldwide MNCS equals 1

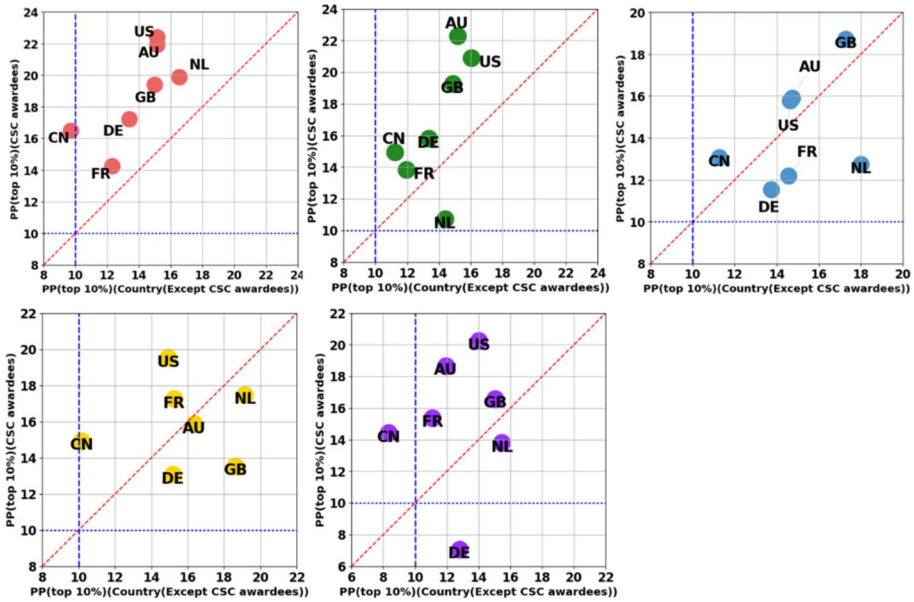

Fig. 4 Proportion of publications by CSC recipients among the top 10% most cited PP (top 10%), compared to those of their home and host countries. *Note*: the reference value for PP (top 10%) is 10%

CSC-funded researchers display partially or overall superior citation performance. For example, publications from the United Kingdom and Germany often receive higher citation counts than those by CSC awardees in the fields of Life and Earth Sciences and Social Sciences and Humanities. In the Netherlands, MNCS values for non-CSC publications are consistently higher than those for CSC awardees, indicating a generally superior citation impact.

MNCS and PP (top 10%) provide complementary perspectives on a country's research impact, capturing both the average citation impact and the concentration of highly influential publications. As shown in Fig. 4, and consistent with the patterns observed in the MNCS, the ability of Chinese scholars to produce high-impact publications is comparatively lower than that of researchers from the other six countries, particularly in the fields of Physical Sciences and Engineering, and Social Sciences and Humanities, where their performance also falls below the global average. By contrast, publications produced by CSC awardees demonstrate a marked improvement in citation impact.

The PP (top 10%) indicator reveals patterns similar to those observed in the MNCS. In the field of Physical Sciences and Engineering, CSC awardees exhibit superior citation performance, consistent with the MNCS results. In the United States, CSC awardees perform comparably to, or even better than, other researchers across all fields. However, in certain fields and countries, publications involving CSC awardees demonstrate lower concentrations of highly cited papers compared to their counterparts. For instance, in the Netherlands, the PP (top 10%) of publications without CSC awardees surpasses that of those involving CSC awardees.

Collaboration patterns

Participation in the CSC programme entails researchers moving abroad, thereby fostering connections with diverse academic communities and expanding international collaboration networks (Zhao et al., 2022). This analysis focuses on the collaboration patterns reflected in the scientific publications of CSC-funded researchers, assessing the extent to which these researchers engage with peers in host countries and beyond—or, conversely, whether they tend to conduct research in relative isolation. Our findings show that CSC-funded publications are, on average, authored by a greater number of researchers compared to the global average, as shown in Fig. 5a. In most fields, these publications also exhibit a higher average number of authors than the broader body of Chinese research, except in the Physical Sciences and Engineering. This elevated average number of co-authors per publication, relative to both global and Chinese benchmarks, suggests a higher degree of collaboration among CSC-funded researchers working abroad.

Figure 5b also illustrates that CSC-funded publications are frequently produced through international collaboration. Such collaborations are particularly prominent among CSC-funded outputs, with nearly 70% involving international co-authorship, substantially higher than the corresponding figures for China and the world, which stand at around 20% in both cases. Given this exceptionally high percentage of international collaborations, we examined whether it might be attributed to the use of multiple affiliations—that is, instances where CSC awardees list both a host country and a Chinese affiliation in the same publication. Our findings reveal that only around 2% of international collaborations can be attributed to the use of multiple affiliations.

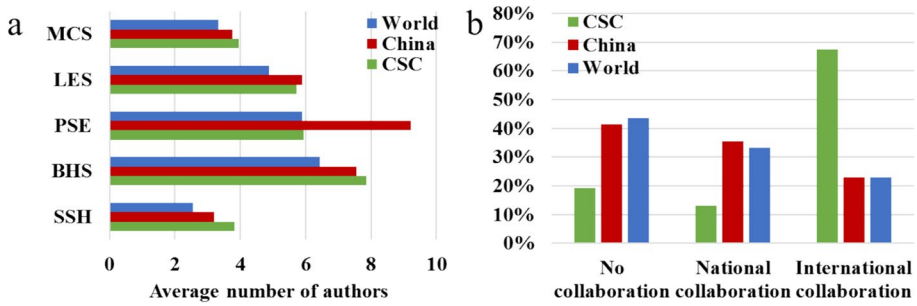

Fig. 5 Comparative collaboration patterns of global, Chinese and CSC-funded publications. **a** Average number of authors per article across fields in global, Chinese and CSC-funded publications. **b** Collaboration patterns across global, Chinese and CSC-funded publications (*BHS* Biomedical and Health Sciences, *LES* Life and Earth Sciences, *MCS* Mathematics and Computer Science, *PSE* Physical Sciences and Engineering, *SSH* Social Sciences and Humanities)

Publications funded by the CSC and produced by a single organisation account for 20% of all CSC-funded publications, a significantly lower proportion compared to both Chinese publications overall and global publications, where around 40% are produced by only one organisation. Both the higher average number of authors and the broader collaboration profile suggest that CSC awardees actively collaborate not only with researchers in their host countries but also with those based abroad.

Figure 6a, b depict the comparative collaboration patterns of CSC-funded publications and all publications in host countries across different fields. We observed that the average number of authors of CSC-funded publications in the selected countries is higher than that of publications from the host countries in every discipline, with the exception of Physical Sciences and Engineering. The higher average number of authors indicates that CSC awardees actively participate in collaborative research, including at the individual level. Within Physical Sciences and Engineering, the average number of authors of CSC-funded publications is lower than that of the overall domestic publications in the host countries.

Researchers supported by the CSC in these six countries exhibit collaboration patterns similar to those observed for CSC-funded publications overall. CSC awardees predominantly engage in collaborative work, with a strong emphasis on international collaboration. The percentage of publications produced solely by CSC awardees and other authors affiliated with the same organisation is notably low, particularly in the US (only about 4% of publications). Conversely, the Netherlands shows a higher percentage of non-collaborative publications (around 26%), compared to other countries and CSC-funded publications overall.

The percentage of international co-publications in France, the Netherlands, and Germany (ranging between 60 and 70%) is slightly lower than the overall rate observed for CSC-funded publications (approximately 70%). Conversely, the United Kingdom, Australia, and the United States exhibit higher proportions of international co-publications compared to the general CSC-funded rate, particularly the US, where more than 90% of publications result from international collaborations. This higher percentage in the US may be partially explained by the tendency of temporary visiting CSC awardees (one- or two-year visitors funded by the CSC) to choose the US as their host destination.

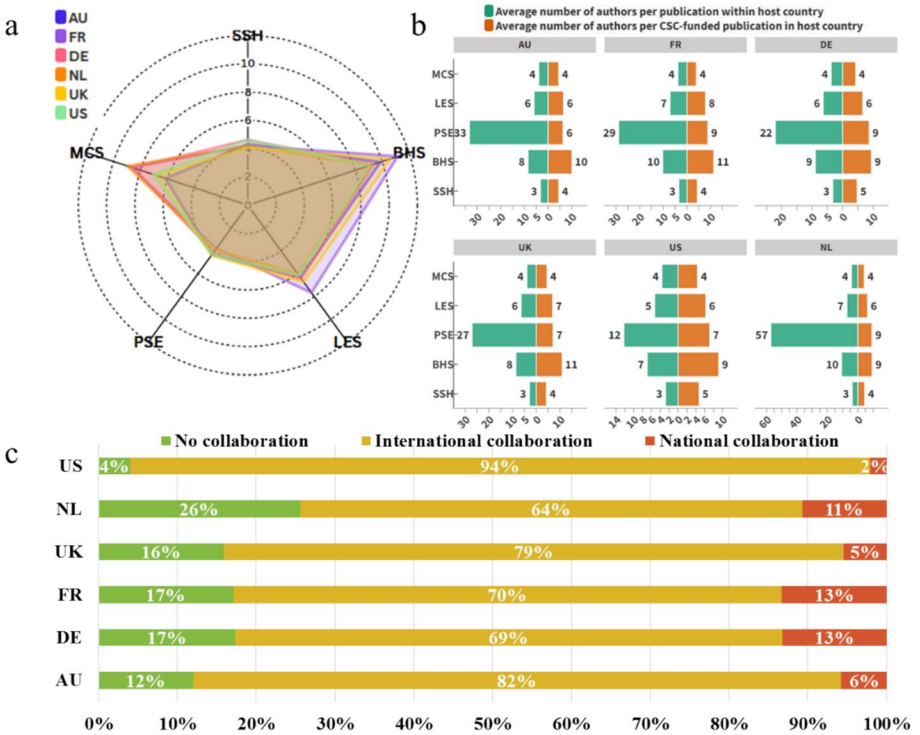

Fig. 6 Comparative collaboration patterns across host countries. **a** Average number of authors per CSC-funded publication by field across host countries. **b** Comparison of the average number of authors per publication between host country publications and CSC-funded publications within those countries. **c** Collaboration patterns of CSC-funded publications across host countries (*BHS* Biomedical and Health Sciences, *LES* Life and Earth Sciences, *MCS* Mathematics and Computer Science, *PSE* Physical Sciences and Engineering, *SSH* Social Sciences and Humanities)

Funding patterns

The CSC programme provides funding for Chinese researchers to undertake research abroad. From this perspective, it may represent an opportunity for host institutions to benefit from the contributions of additional researchers without bearing the associated salary costs. This could potentially support the development of research lines within host institutions that may otherwise struggle to secure funding (Hooghe & Martin, 2024). But is this actually the case? To explore this question, we examine whether CSC-funded scholars are active in areas that tend to be underfunded. Specifically, we analyse the funding acknowledgements in publications authored by CSC-funded researchers to determine whether the CSC is the sole source of financial support or whether other funding sources are also acknowledged. This serves as a rough proxy to assess whether the research areas pursued by CSC-funded scholars are typically well-resourced or not.

For each host country, we then compared the percentage of CSC-funded publications based on the number of funding organisations acknowledged. A first observation is that approximately 40% of these publications do not acknowledge any source of funding.

This proportion varies slightly across countries, ranging from 35% in the United Kingdom to 42% in France.

We also observe that CSC-funded publications tend to acknowledge a higher number of funding organisations compared to other publications from the host countries. This greater number of funders suggests that CSC-funded researchers are often active in areas that are already well-funded. Moreover, the presence of multiple organisations in the funding acknowledgements may also indicate that these researchers are engaged in prominent, well-supported research areas, potentially involving collaboration between several organisations and countries.

Due to potential variations across disciplines, we further explored the role of the CSC within different fields. Table 4 presents the distribution of acknowledged funders in CSC-funded publications compared to other publications produced in the host countries. In the table, red shades indicate a higher percentage of publications acknowledging funding, while blue shades indicate a lower percentage.

As illustrated in Table 4, we observe varying trends in the number of funders acknowledged across different disciplines. The Social Sciences and Humanities show the highest proportion of publications without funding acknowledgements (approximately 70%). Similarly, Mathematics and Computer Science also display a relatively high percentage of publications lacking funding acknowledgements (50–60%). Furthermore, in line with the funding patterns identified in all publications from the host countries (Fig. 7), the number of funders acknowledged across disciplines is generally concentrated around zero or one. Hosting CSC-funded researchers may therefore be particularly beneficial in fields such as Social Sciences and Humanities and Mathematics and Computer Science, given the substantial proportion of publications produced in these areas that receive little or no external funding.

Examining CSC-funded publications across disciplines, we observe that the pattern shown in Table 4 is consistent with Fig. 7, that is, CSC-funded publications tend to acknowledge a higher number of funders than the overall set of publications from host countries, particularly in seemingly well-funded disciplines such as Physical Sciences and Engineering and Life and Earth Sciences. Notably, a striking pattern emerges in disciplines where funding acknowledgements are less common, particularly in the Social Sciences and Humanities. In these fields, the CSC frequently appears as the sole funder for a substantial proportion of publications. In Australia, the United Kingdom, and the Netherlands, the share of CSC-only funded publications exceeds 40%, with the Netherlands displaying the highest proportion, at over 50%. By contrast, in Germany, France, and the United States, the proportion of publications exclusively funded by the CSC is comparatively lower, at around one-third. This substantial presence of CSC as the sole funding source may suggest that the programme may play a particularly important role in supporting research in the Social sciences and humanities, where other sources of funding appear to be more limited.

Connections to the "Seven Sons of National Defence" universities

One aspect that has come under scrutiny in recent years regarding the CSC programme is the relationship between some CSC-funded researchers and a group of universities commonly referred to as the "Seven Sons of National Defence." This group comprises seven Chinese universities known for their close ties to the People's Liberation Army. These connections have raised concerns about potential risks for countries hosting CSC-funded scholars, with arguments suggesting that the Chinese military might benefit from research

Table 4 Percentage of publications by number of acknowledged funders breakdown by disciplines and host countries

		Percentage of funders for publications by fields within host country						Percentage of funders for CSC-funded publications by fields within host countries					
		0	1	2	3	4	>4	0	1	2	3	4	>4
UK	BHS	30%	19%	14%	10%	7%	20%	0%	15%	21%	22%	17%	26%
	LES	20%	19%	18%	14%	10%	19%	0%	15%	30%	24%	17%	14%
	MCS	44%	19%	15%	10%	6%	7%	0%	21%	31%	22%	17%	9%
	PSE	22%	16%	17%	14%	10%	21%	0%	16%	29%	26%	15%	15%
	SSH	67%	13%	10%	4%	2%	3%	0%	41%	26%	15%	8%	10%
US	BHS	32%	25%	16%	10%	6%	11%	0%	8%	34%	29%	16%	13%
	LES	21%	22%	20%	14%	9%	14%	0%	12%	32%	27%	17%	12%
	MCS	48%	19%	14%	9%	5%	6%	0%	15%	29%	30%	15%	13%
	PSE	24%	21%	19%	13%	9%	15%	0%	10%	27%	28%	18%	17%
	SSH	73%	13%	7%	3%	2%	2%	0%	25%	31%	23%	10%	10%
AU	BHS	35%	21%	14%	9%	6%	14%	0%	15%	36%	26%	13%	9%
	LES	23%	23%	20%	14%	8%	13%	0%	16%	33%	28%	15%	9%
	MCS	50%	18%	13%	8%	5%	6%	0%	24%	32%	25%	11%	8%
	PSE	25%	20%	18%	13%	8%	15%	0%	12%	37%	29%	14%	8%
	SSH	70%	15%	7%	3%	2%	2%	0%	42%	26%	22%	8%	3%
NL	BHS	35%	24%	13%	8%	5%	14%	0%	37%	29%	18%	8%	8%
	LES	25%	25%	18%	12%	7%	13%	0%	41%	28%	17%	8%	6%
	MCS	53%	21%	12%	6%	3%	4%	0%	50%	26%	15%	7%	2%
	PSE	26%	21%	16%	11%	7%	19%	0%	41%	30%	17%	7%	5%
	SSH	67%	17%	8%	4%	2%	2%	0%	52%	24%	14%	6%	4%
FR	BHS	42%	16%	11%	8%	6%	17%	0%	17%	18%	21%	13%	31%
	LES	24%	20%	18%	13%	9%	16%	0%	29%	25%	21%	12%	14%
	MCS	57%	17%	12%	6%	4%	4%	0%	45%	26%	17%	7%	5%
	PSE	34%	19%	15%	10%	7%	15%	0%	31%	24%	19%	13%	13%
	SSH	72%	11%	7%	4%	2%	4%	0%	28%	28%	24%	12%	8%
DE	BHS	39%	22%	13%	8%	5%	12%	0%	26%	34%	21%	10%	10%
	LES	24%	27%	18%	12%	7%	12%	0%	28%	31%	23%	10%	7%
	MCS	54%	22%	12%	6%	3%	3%	0%	33%	27%	21%	12%	8%
	PSE	28%	24%	17%	11%	7%	14%	0%	27%	29%	21%	12%	11%
	SSH	68%	18%	8%	3%	2%	2%	0%	32%	34%	22%	4%	7%

BHS Biomedical and Health Sciences, *LES* Life and Earth Sciences, *MCS* Mathematics and Computer Science, *PSE* Physical Sciences and Engineering, *SSH* Social Sciences and Humanities

conducted in sensitive areas. Consequently, such links between CSC awardees and host institutions are sometimes viewed as possible unwanted foreign government interference and a potential future security challenge for the host countries. For example, the US Department of Defence has blacklisted certain foreign research organisations, including those universities within the "Seven Sons of National Defence," to prevent problematic collaborations.

As illustrated in Fig. 8, our analysis shows that only a small proportion of CSC-funded publications (approximately 5%) areproduced by researchers affiliated with the "Seven Sons of National Defence" universities. However, in countries such as the UKand France, the relatively high percentages observed in the early years are largely attributable to the low overall number of CSCfundedpublications during that period. Notably, the proportion of publications involving any of the universities within the "SevenSons of National Defence" group has declined over time, reaching 2.6% in 2021.

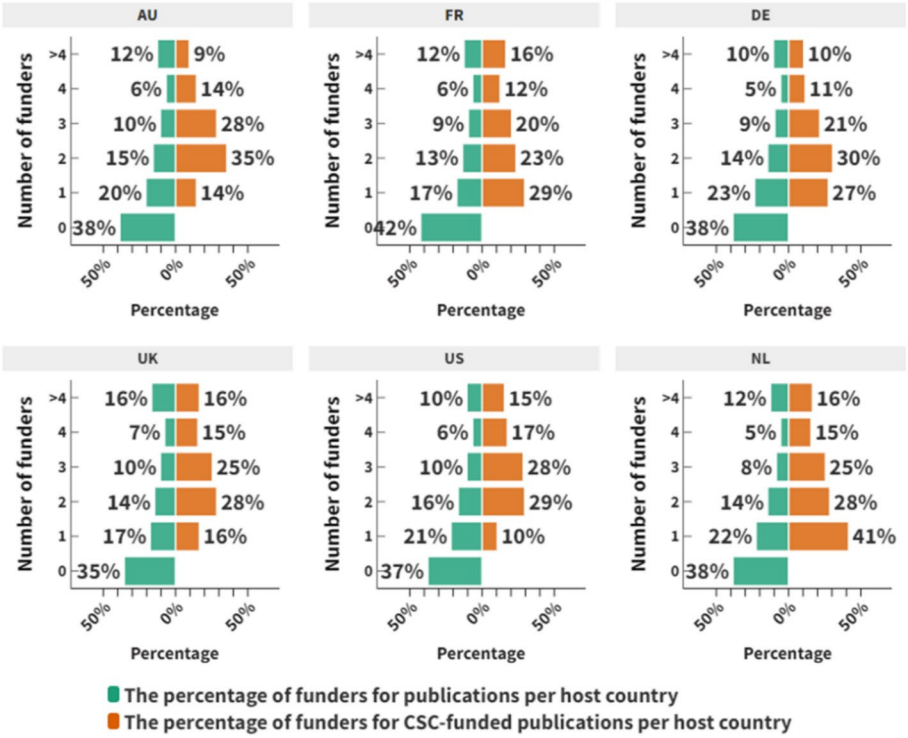

Fig. 7 Percentage of publications by number of acknowledged funders (ranging from 0 to 4 or more) across host countries: comparison between all publications and CSC-funded publications

The US consistently maintained a relatively higher percentage of publications involving the "Seven Sons of National Defence" compared to the overall CSC-funded papers. This percentage peaked at 9.4% in 2013 and subsequently stabilised between 4.8 and 6.8%. The initially high percentages observed in the UK and France declined markedly over the following years, dropping to levels even lower than the overall proportion of CSC-funded publications. Meanwhile, Germany and the Netherlands exhibited comparatively lower and more stable percentages throughout the period.

These findings suggest that only a limited number of CSC awardees have links with the "Seven Sons of National Defence" universities, at least during their time abroad supported by the CSC. Even in the US, the highest proportion of CSC-funded publications co-authored with these universities remains below 10%.

Dual-use research activities by CSC awardees

Dual-use technology is frequently cited as one of the greatest risks in research collaborations with certain countries (Vaynman & Volpe, 2023), and it is an increasing concern among host countries of CSC-funded researchers. Our results indicate that, overall, the proportion of publications related to dual-use research is low (around 0.5%) among CSC-funded publications, both within China and globally, as shown in Table 5. This suggests

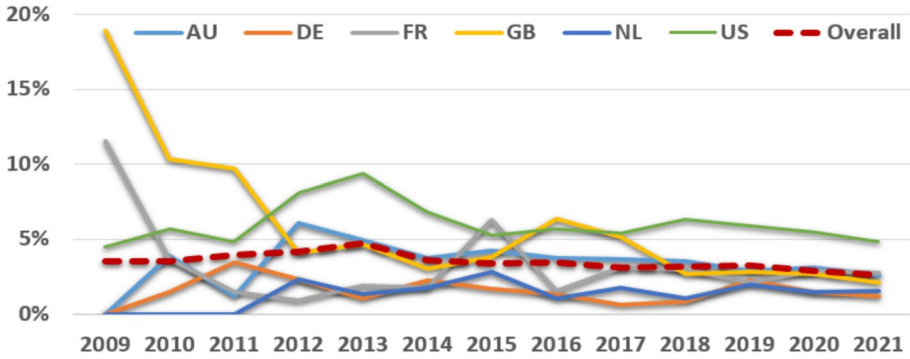

Fig. 8 Percentage of CSC-funded publications co-authored with the "Seven Sons of National Defence" universities across host countries

that the majority of research conducted worldwide is not related to dual-use applications, a pattern that holds for China and CSC-funded research as well.

Figure 9 illustrates the variation in the percentage of publications related to dual-use research across different categories. Examining these differences offers insights into the dual-use research focus of CSC-funded publications, host countries, and the contribution of CSC-funded researchers by category. CSC-funded publications show a stronger emphasis than both China and global publications in several categories, including Marine (Cat. 8) and Aerospace and Propulsion (Cat. 9). Conversely, the global corpus places a greater focus on Pathogens (Cat. 1), with a notably higher percentage compared to China and CSC awardees. Overall, the share of dual-use research publications remains limited, with categories such as Computers (Cat. 4) consistently showing the lowest levels of emphasis across all contexts.

As illustrated in Fig. 10, the focus on dual-use research across most host countries follows a distribution similar to that of worldwide publications, except countries that are more active in Pathogens (Cat. 1). This similarity in distribution patterns indicates comparable levels of dual-use research activity across categories between the global set and individual host countries.

However, notable differences exist between the dual-use research publications of host countries and those by CSC-funded researchers. CSC awardees tend to concentrate on specific categories, in contrast to the broader dual-use research portfolio of their host countries. For example, CSC-funded researchers in France and Germany place greater emphasis on Electronics (Cat. 3). In the UK and the US, CSC awardees are more active in Special Materials and Related Equipment (Cat. 1-1) compared to both global trends and their host country peers. Meanwhile, CSC researchers in the Netherlands and Australia focus more on Telecommunications & Information Security (Cat. 5), Sensors and Lasers (Cat.

Table 5 Number and percentage of dual-use research publications

	Dual-use research publications	All publications	Percentage of dual-use research publications (%)
World	114,887	25,338,451	0.45
China	27,237	5,113,438	0.53
CSC	562	104,148	0.54

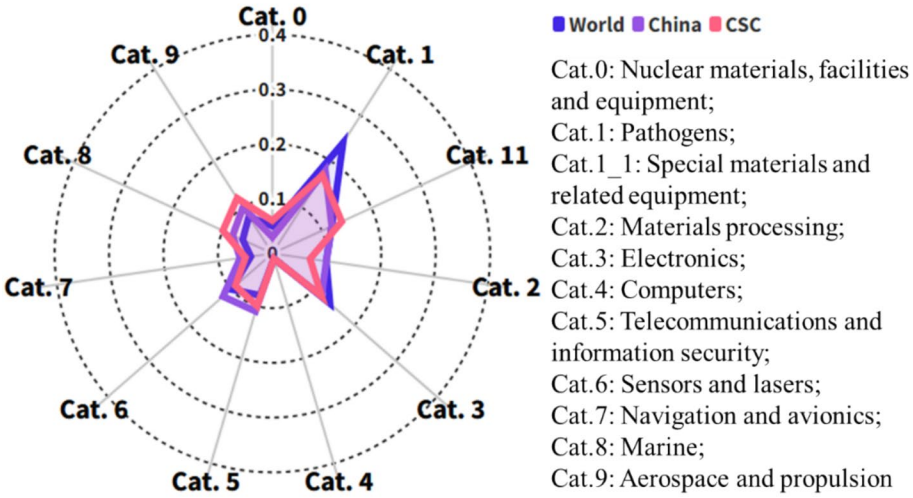

Fig. 9 Percentage of CSC-funded publications across dual-use categories

6), and Marine (Cat. 8). Overall, this suggests that the CSC programme may prioritize specific technological fields that differ from the general dual-use research focus of the host countries.

Discussion and conclusion

International researcher mobility is generally regarded as a reciprocal process that benefits both home and host countries. However, geopolitical tensions and mobility-related challenges have raised questions about these mutual benefits. Recently, several countries hosting Chinese researchers funded by the CSC have expressed security concerns. These concerns focus mainly on the training of Chinese students in sensitive research areas, such as dual-use technologies, and the involvement of scholars affiliated with the "Seven Sons of National Defence," a group of universities closely linked to the Chinese military. Such issues are viewed as potential threats to the strategic security of host countries. Motivated by these concerns, we analysed the potential benefits and challenges of the CSC programme across five dimensions: publication output, citation impact, collaboration patterns, funding profiles, and security-sensitive research. Regarding security-sensitive research, our focus was primarily on CSC-funded publications co-authored by "Seven Sons of National Defence" scholars and those involving dual-use research.

This study contributes to the literature on the evaluation of mobility funding programmes by focusing on both home and host countries. To our knowledge, it is the first attempt to assess government-sponsored international mobility programmes from the perspective of both home and host nations. Our findings also provide valuable evidence for policymakers by addressing some of the concerns surrounding the CSC programme. This study offers important insights and implications for international mobility supported by the CSC, particularly as the current heightened scrutiny of CSC awardees may disrupt existing networks and potentially have

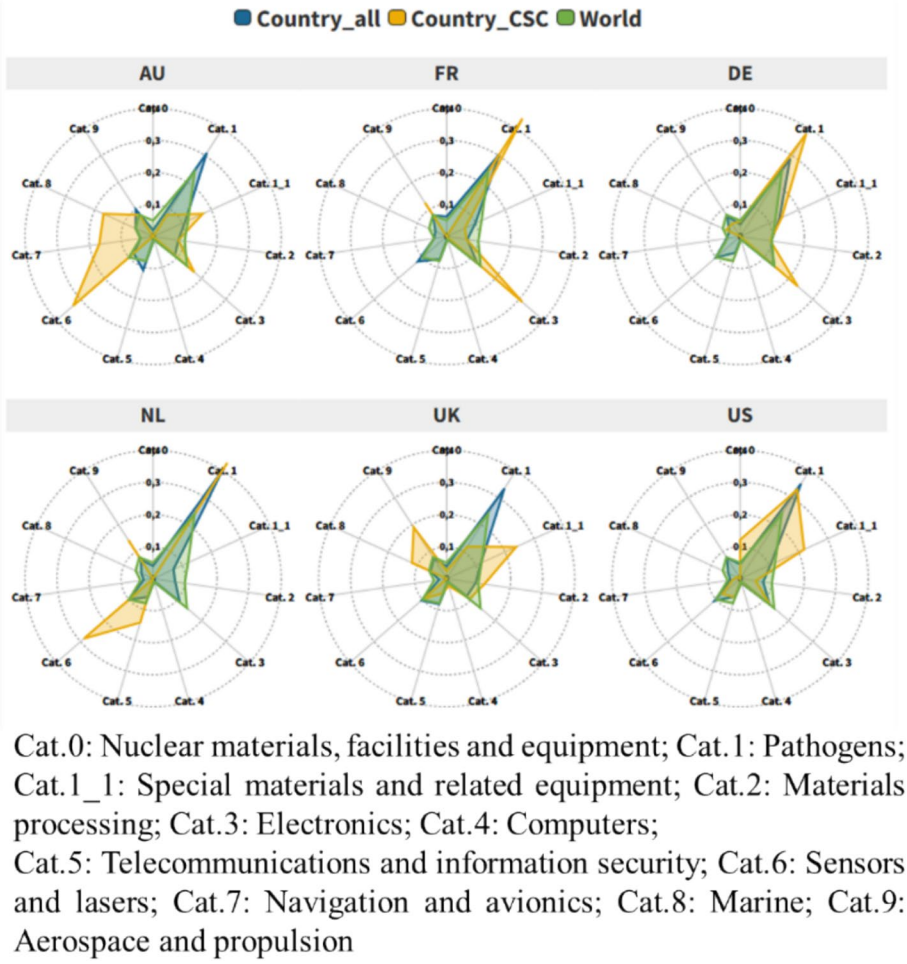

Fig. 10 Proportion of dual-use research by category in host countries and CSC-funded publications

adverse effects on numerous countries, including major science producers such as the United States (Sugimoto et al., 2017).

Regarding our first research question, which concerned the research profile of CSC-funded researchers across disciplines, we observed that CSC-funded publications are primarily concentrated in the fields of Physical Sciences and Engineering, as well as Life and Earth Sciences, aligning with China's national strategic development goals for the period 2006–2020 (State Council of the People's Republic of China, 2006; Wang et al., 2024). Additionally, CSC awardees tend to engage in disciplines where host countries demonstrate significant academic strengths. Thus, both national policy priorities and host-country research strengths shape the mobility patterns of CSC awardees. In this regard, the mobility patterns we observe reflect broader global dynamics, where researchers are drawn to leading scientific hubs (Horta et al., 2020).

With respect to our second research question, on the citation impact of CSC-funded researchers, we found that their publications demonstrated a higher mean normalised citation impact

and a greater proportion of top 10% most-cited publications compared to both China and global benchmarks. Wang et al. (2024) also found that CSC-funded publications outperformed the China mainland average and far exceeded the worldwide average. Our study supports the positive association between international mobility and citation impact (Kotsemir et al., 2022; Song & Gan, 2022). This outperformance in citation impact can be partially attributed to enhanced access to knowledge, advanced scientific infrastructure, and higher-quality collaboration networks (Baruffaldi et al., 2020; Gu et al., 2024; Shen, 2018). However, citation impact varies by host country and discipline. For instance, CSC awardees within Physical Sciences and Engineering outperform their domestic counterparts, while those in Life and Earth Sciences and Biomedical and Health Sciences lag behind their host-country peers.

Turning to the third research question, related to the collaboration patterns of CSC-funded researchers, we observed that their publications have a higher average number of authors and a greater proportion of international collaborations compared to publications worldwide, in China, and host countries. The elevated average number of authors and higher rate of international collaborations suggest that CSC-funded researchers are well embedded within collaborative networks both within host countries and internationally. These findings align with previous studies (e.g., Gu et al., 2024; Horta et al., 2021; Wang et al., 2024), which also highlighted the role of the CSC programme in facilitating the integration of Chinese researchers into global scientific collaboration networks.

Our fourth research question examined whether CSC-funded researchers tend to work in relatively underfunded areas. Our results reveal that CSC-funded publications tend to acknowledge a greater number of funders than other publications from host countries. This suggests that CSC-funded researchers often engage in research areas that are already well supported, potentially due to the collaborative nature of these fields, which increases the likelihood of attracting multiple funding sources. However, in areas such as the Social Sciences and Humanities, where funding acknowledgements are frequently absent or minimal, CSC support may provide valuable resources that would otherwise be difficult to obtain. Similarly, in disciplines like Mathematics and Computer Science in some European countries (e.g., France, the Netherlands, and Germany), the CSC is often the sole acknowledged funder. These findings support the view that the CSC programme enables researchers, particularly in underfunded fields, to undertake research that might otherwise struggle to secure financial support (Hooghe & Martin, 2024).

Our final research question considered whether CSC-funded researchers are typically active in sensitive areas. Our findings indicate a low proportion of CSC awardees affiliated with the "Seven Sons of National Defence" universities, at least during their overseas period and in publications produced abroad. The small number of publications mentioning these universities suggests limited direct engagement of CSC awardees with institutions associated with the military. However, these findings are based on the institutional affiliations provided by authors in their publications, assuming that all authors accurately report their affiliations. We cannot guarantee that all affiliations are consistently included or that some may not have been intentionally omitted, which could potentially affect the results. Furthermore, affiliations established after the mobility period fall outside the scope of this study. Our results also show that dual-use research is not a significant focus among CSC-funded publications. This finding may help alleviate current concerns about potential future challenges for host countries. Nonetheless, these conclusions remain tentative due to limitations in identifying dual-use content and the possibility that sensitive research may not be reflected in academic publications.

Limitation and future research

There are limitations to this study, as it only considers CSC-funded publications indexed in WoS, thereby excluding other types of scientific outputs such as patents, books, as well as any other research activities not captured in WoS records. Additionally, WoS has limited coverage in certain fields, notably the Social Sciences and Humanities. Regarding funding information, beyond some technical issues affecting the accurate processing of funding acknowledgements in databases (Paul-Hus et al., 2016), there may also be instances where researchers fail to acknowledge funding as required.

Future research could address several important issues. One key avenue involves examining CSC-funded international mobility programmes in greater detail. As discussed in "Research background" section, these programmes encompass various funding schemes, each with distinct objectives, priorities and structures. Prior research has demonstrated heterogeneity in mobility duration (Gureyev et al., 2020). Consequently, further studies could consider the variety of funding schemes and focus on specific initiatives, such as the *National Regional Studies and High-Level Foreign Language Talent Development Programme*, to explore the differential effects of mobility duration. Additionally, the causal relationship between mobility supported by CSC and its impact on knowledge production and collaboration is another interesting research direction.

Finally, while our study assessed the payoffs and challenges of CSC funding programmes at the national level for host countries, a more detailed analysis at the level of host organisations would be valuable. Such an analysis could yield deeper insights into how specific host institutions benefit from and adapt to the arrival of Chinese researchers.

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Author contributions Qianqian Xie: Conceptualization; Data curation; Formal Analysis; Investigation; Methodology; Software; Visualization; Writing—original draft. Alfredo Yegros: Conceptualization; Data curation; Formal Analysis; Investigation; Methodology; Visualization; Supervision; Writing—review & editing.

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Data availability The data used in this study have been deposited in Zenodo: https://doi.org/10.5281/zenodo.14105510.

Code availability The code used in this study has been deposited in GitHub**:** https://github.com/Qianqian727/Matching-authors-name-in-sentences.git.

Declarations

Conflict of interest One of the authors is funded by the CSC scholarship.

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